2 Program Logic: Arithmetic Calculations and Operators, Selection

Animal Distribution

Create a program named **animal_distribution** that first asks the user to input the number of dogs, the number of cats, and the number of parrots. Then the program should calculate the *percentage of each animal species*. The program should display the percentages with one decimal place.

Example output:

```
Enter the number of female students: 600
Enter the number of male students: 200

Female: 75.0 %
Male: 25.0 %
```

Sales Bonus

Create a program named **sales_bonus** that first inputs the selling price of a boat from the user. Then the program should calculate the salesperson's bonus with two decimals.

The sales bonus is 1 % of the selling price if the boat costs less than 80000. Otherwise, the bonus is 1,75 % of the selling price. The minimum bonus is 400 euros.

For example, if the selling price is 25000 euros, then the bonus is 400 euros. If the selling price is 60000 euros, the bonus is 600 euros. If the selling price is 100000 euros, the bonus is 1750 euros.

Example output:

```
Enter the car's selling price: 60000
The bonus is 900.00 euros.
```

Triathlete Program

Create a program named **triathlete_program** that first asks the user to input the number of energy bars and the number of triathletes. Then the program should compute how many energy bars each triathlete receives and how many energy bars are left over. Each triathlete must receive the same number of bars. You can use the remainder operator to determine the number of leftover energy bars.

Example output:

```
How many apples? 14
How many children? 4

Each child will get 3 apples.
There will be 2 leftover apples.
```

Guitar Strings

Create a program named **guitar_strings** that calculates the total cost of guitar strings a guitarist will use on a set of gigs.

The program first inputs the *total number of gigs the guitarist will do* (integer), the *number of gigs the guitarist plays with the same set of strings*, and the *price of one set of guitar strings*. Then the program should compute and print how many sets of guitar strings the guitarist will need and the total cost of guitar strings, formatted to two decimal places.

This time, assume that the guitarist *must* buy new guitar strings if he is going to play any gigs.

Example output:

```
Number of gigs: 8
Gigs to be played with the same set of strings: 3
Price of a set of guitar strings: 7.50

The guitarist needs 3 new sets of guitar strings
The total cost is 22.50 euros
```

NB! Please note that Python has two division operators that behave differently.

Ticket Prices

Create a program named **ticket_prices** that first inputs the number of days when the user visits the local gym per year, the yearly membership fee, and the price of the day pass. Then, the program should determine which one is the cheaper option (paying the yearly membership fee or buying separate day passes) and how much cheaper it is.

Example output:

```
Enter the number of days of gym visits per year: 65
Enter price for a day pass: 12.00
Enter yearly membership fee: 650

Paying the yearly membership fee is 130.00 euros cheaper
```

The program should display one of the following texts:

Buying day passes is 99.00 euros cheaper

Paying the yearly membership fee is 99.00 euros cheaper

There is no price difference

Acurate Arithmetic

Create a program named **accurate_arithmetic** that first inputs two decimal numbers from the user. Then the program should print the sum of the decimal numbers as shown in the example output. Note: You are not required to format the number of decimal places in the output.

Note

- Please do not be too surprised, if your code fails a couple of times in Viope testing before you have finished it.
- Please study the course materials carefully to see how you can perform ***exact computations*** on decimal values.

Example output:

```
Enter first number: 1.3
Enter second number: 2.55

1.3 + 2.55 = 3.85
```

Example output:

```
Enter first number: 123456789.0
Enter second number: 0.987654321

123456789.0 + 0.987654321 = 123456789.987654321
```